

Project Title
(Mini Search Engine / Information-Retrieval System)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project presents the design and implementation of a Mini Search Engine / Information Retrieval System that efficiently indexes a collection of documents and supports four types of search queries: exact keyword search, phrase/substring search, fuzzy ("Did You Mean") search, and document similarity search. The system employs an inverted index for fast keyword retrieval, positional indexing for phrase matching, edit distance algorithms for spelling correction, and TF-IDF vector representation with cosine similarity for identifying related documents. Each algorithm is implemented from scratch without relying on built-in search libraries, enabling a deeper understanding of information retrieval principles. The search engine ranks results based on relevance, ensuring accurate and meaningful responses to user queries. The project also analyzes the time and space complexity of every core algorithm to justify its efficiency and scalability. This implementation demonstrates fundamental search engine concepts while providing a practical, lightweight, and extensible platform for learning modern information retrieval techniques.

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